

Name: Solutions

These questions assess your understanding of the material from Chapters 7, 8 of OpenStax College Physics. Please write your name on the sheets of paper that you use and show all your work. *full credit is awarded only for complete work.*

i.e. no points are awarded if complete work is not shown.

1. James, a 80-kg ice-hockey goalie who is originally at rest, catches a 0.15-kg ice-hockey puck that is traveling at a velocity of 42.6 m/s. What is the net change in James's and the puck's kinetic energy during the collision? Assume friction is negligible.

find ΔKE where $KE = \frac{1}{2}mv^2$

$$\Delta X = X_F - X_i$$

$$\Delta KE = KE_F - KE_i$$

ANSWER: -135.85 J

what type of collision is this? "catches" $\therefore$ perfectly inelastic $\therefore \Delta \vec{p} = 0$ & colliding objects stick together



before collision

$m_p = 0.15 \text{ kg}$
 $v_p = 42.6 \text{ m/s}$
 $m_J = 80 \text{ kg}$
 $v_J = 0$

after collision

$m' = m_J + m_p$
 $m' = 80 \text{ kg} + 0.15 \text{ kg}$
 $m' = 80.15 \text{ kg}$
 $v' = ?$

$\Delta \vec{p}_x = 0$
 $\vec{p}_{fx} - \vec{p}_{ix} = 0$
 $\vec{p}_{fx} = \vec{p}_{ix}$
 $m'v' = m_p v_p + m_J v_J$
 $v' = \frac{1}{m'} (m_p v_p + m_J v_J)$
 $v' = \frac{1}{m'} (m_p v_p) \because v_J = 0$

(p_F is also known as p' since p_{fx} is more cumbersome than p'_x)

find ΔKE .

$$\Delta KE = KE_F - KE_i$$

$$\Delta KE = \frac{1}{2} m' (v')^2 - \left(\frac{1}{2} m_p v_p^2 + \frac{1}{2} m_J v_J^2 \right)$$

$$\Delta KE = \frac{1}{2} m' \left(\frac{m_p v_p}{m'} \right)^2 - \frac{1}{2} m_p v_p^2 \quad (v_J = 0)$$

$$\Delta KE = \frac{1}{2} m_p v_p^2 \left(\frac{m_p}{m'} - 1 \right)$$

$$\Delta KE = \frac{1}{2} (0.15 \text{ kg}) (42.6 \text{ m/s})^2 \left[\frac{0.15 \text{ kg}}{80.15 \text{ kg}} - 1 \right]$$

$$\Delta KE = -135.85 \text{ J}$$

2. Victoria normally requires 11400 kJ (about 2724 Calories) of food energy per day. If Victoria consumes 12084 kJ per day, she will steadily gain weight. How much time must Victoria spend bicycling per day, if she does not want to gain weight? Victoria's average power output while bicycling is 360 W. Report your answer in minutes.

ANSWER: 31.67 min

$$P = \frac{W}{t}$$

She must do (12084 kJ - 11400 kJ) = 684 kJ of work to maintain her weight

$$t = \frac{W}{P}$$

$$t = \frac{684 \text{ kJ} \left| \frac{1000 \text{ J}}{1 \text{ kJ}} \right|}{\frac{360 \text{ J}}{1 \text{ s}}} = 1900 \text{ s} \left| \frac{1 \text{ min}}{60 \text{ s}} \right| = \underline{31.67 \text{ minutes}}$$

3. Joshua, a 99-kg ice-hockey goalie who is originally at rest, is hit by a 0.19-kg ice-hockey puck that is traveling at a velocity of 36.1 m/s. Joshua hits the puck in such a way that this may be considered an elastic collision. Calculate the speed of the goalie after he hits the puck.

find v'_j
 v'_p is unknown

for elastic collision, $\Delta \vec{p} = 0$ & $\Delta KE = 0$

ANSWER: 13.83 m/s

$$\Delta \vec{p} = 0$$

$$\vec{p}' - \vec{p} = 0$$

$$\vec{p}' = \vec{p}$$

$$m_j \vec{v}'_j + m_p \vec{v}'_p = m_p \vec{v}_p + m_j \vec{v}_j$$

$$m_j v'_j - m_p v_p = m_p v_p \quad (v_j = 0)$$

what is v'_p ?

$$m_p v'_p = m_j v'_j - m_p v_p$$

$$v'_p = \frac{1}{m_p} [m_j v'_j - m_p v_p]$$

$$\therefore v_p^2 = \frac{1}{m_p^2} [m_j^2 v_j'^2 - 2m_j v'_j m_p v_p + m_p^2 v_p^2]$$

$$\Delta KE = 0$$

$$KE' - KE = 0$$

$$KE' = KE$$

$$\frac{1}{2} m_p v_p'^2 + \frac{1}{2} m_j v_j'^2 = \frac{1}{2} m_p v_p^2 + \frac{1}{2} m_j v_j^2$$

$$\frac{1}{2} m_p v_p'^2 + \frac{1}{2} m_j v_j'^2 = \frac{1}{2} m_p v_p^2 \quad (\because v_j = 0)$$

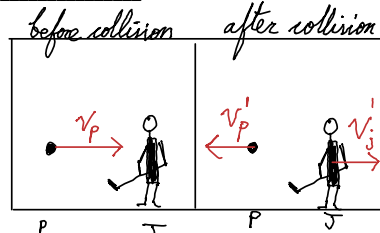
$$m_p v_p'^2 + m_j v_j'^2 = m_p v_p^2$$

$$v_p'^2 = \frac{1}{m_p} [m_p v_p^2 - m_j v_j'^2]$$

equate these.

$$\left(\vec{v}'_p = -v'_p \right)$$

direction



Solve for v'_j

$$\frac{1}{m_p^2} [m_j^2 v_j'^2 - 2m_j v'_j m_p v_p + m_p^2 v_p^2] = \frac{1}{m_p} [m_p v_p^2 - m_j v_j'^2]$$

$$m_j^2 v_j'^2 - 2m_j v'_j m_p v_p + m_p^2 v_p^2 = m_p (m_p v_p^2 - m_j v_j'^2)$$

$$(m_j^2 + m_p m_j) v_j'^2 - (2m_p m_j v_p) v'_j = 0$$

$$(v_j' \neq 0) \therefore (m_j^2 + m_p m_j) v'_j - 2m_p m_j v_p = 0$$

$$v'_j = \frac{2m_p m_j v_p}{m_j(m_j + m_p)} = \frac{2m_p v_p}{m_p + m_j} = \frac{2(0.19 \text{ kg})(36.1 \text{ m/s})}{0.19 \text{ kg} + 99 \text{ kg}} = \underline{0.1383 \text{ m/s}}$$

4. Hannah is playing baseball and is sliding into second base. She has a mass of 85.8 kg and begins sliding at a speed of 7.9 m/s. Measurements show that she covers a distance of 5.1 m during her slide. What is the magnitude of the average friction force acting against Hannah during her slide?

ANSWER: 524 N

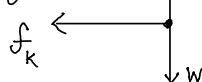


find f_k

Work-energy theorem: $W_{net} = \Delta KE = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_0^2$

definition of work: $W_{net} = |\vec{F}_{net}|d \cos \theta_{Fd}$

free-body diagram:



$\vec{F}_{net y} = 0 \therefore a_y = 0 \therefore N = mg$

$\vec{F}_{net x} = -f_k$

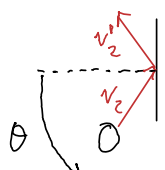
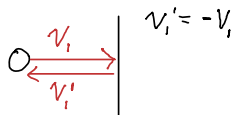
$$\begin{aligned} W_{net} &= W_{net} \\ \frac{1}{2}mv_f^2 - \frac{1}{2}mv_0^2 &= |\vec{F}_{net}|d \cos \theta_{Fd} \\ -\frac{1}{2}mv_0^2 &= f_k d \cos 180^\circ \\ \frac{1}{2}mv_0^2 &= f_k d \end{aligned}$$

$$f_k = \frac{mv_0^2}{2d} = \frac{(85.8 \text{ kg})(7.9 \frac{\text{m}}{\text{s}})^2}{2(5.1 \text{ m})}$$

$f_k = 524 \text{ N}$

5. Two identical billiard balls strike a rigid wall with the same speed and bounce without any change of speed. The first ball strikes perpendicular to the wall. The second ball strikes the wall at an angle of 80.5° from the perpendicular, and bounces off at an angle of 80.5° from perpendicular to the wall. Calculate the ratio of the magnitudes of impulses on the two balls from their collision with the wall.

ANSWER: 6.06



impulse: $\Delta \vec{p}$

find $\frac{\Delta p_1}{\Delta p_2}$

$$\frac{\Delta p_1}{\Delta p_2} = \frac{2mv}{2mv \cos \theta} = \frac{1}{\cos \theta} = \frac{1}{\cos 80.5^\circ}$$

$\frac{\Delta p_1}{\Delta p_2} = 6.0588$

$$|\Delta \vec{p}_1| = |\vec{p}_1' - \vec{p}_1|$$

$$\Delta p_1 = |m v_i' - m v_i|$$

$$\Delta p_1 = |m(-v_i) - m v_i|$$

$$\Delta p_1 = |-2m v_i| = 2m v_i$$

$$\Delta p_1 = 2m v$$

$$|\Delta \vec{p}_2| = \sqrt{\Delta p_{2x}^2 + \Delta p_{2y}^2}$$

$$\Delta p_{2x} = |m v_{ix}' - m v_{ix}| = |m(-v_x) - m v_x|$$

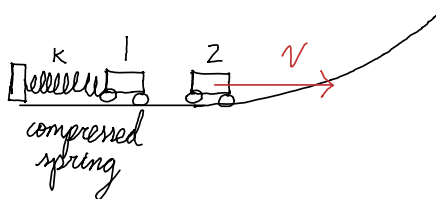
$$\Delta p_{2x} = |-2m v_x| = 2m v_x = 2m v \cos \theta$$

$$\Delta p_{2y} = |m v_{iy}' - m v_{iy}| = 0$$

$$\Delta p_2 = \sqrt{\Delta p_{2x}^2 + 0^2} = \Delta p_{2x} = 2m v \cos \theta$$

6. Kevin places a 0.12-kg toy car next to a spring on a track. He moves the car backward to compress the spring 9.1 cm, then releases the car, propelling it upward along a curved track. How fast is the car going after it is released, before it begins travelling up the slope? Since Kevin greased the axles of the toy car, assume there is no friction. Kevin's previous experiments indicate that the spring has a force constant of 190 N/m.

ANSWER: 3.62 m/s



no friction
no externally applied forces.
 $\therefore$ no nonconservative forces
 $\therefore W_{nc} = 0$
 $\therefore \Delta E_t = 0$ ($\because \Delta E_t = W_{nc}$)

find v

$$\Delta E_t = 0$$

$$E_2 - E_1 = 0$$

$$E_2 = E_1$$

$$\frac{1}{2}mv^2 = \frac{1}{2}k(\Delta x)^2$$

$$v = \Delta x \sqrt{\frac{k}{m}}$$

$$v = (9.1 \times 10^{-2} \text{ m}) \sqrt{\frac{190 \frac{\text{N}}{\text{m}}}{0.12 \text{ kg}}}$$

$v = 3.62 \text{ m/s}$

(on units: $\frac{\text{N}}{\text{m}} = \frac{\text{kgm}}{\text{s}^2} = \frac{\text{kg}}{\text{s}^2}$)
 $\therefore \sqrt{\frac{\text{N/m}}{\text{kg}}} = \sqrt{\frac{1}{\text{s}^2}} = \frac{1}{\text{s}}$

7. Andrew steps off a platform onto the floor from a height of 3.1 m. If Andrew has a mass of 96.6 kg and lands without bending his knees, his knee joints compress by 0.49 cm. Calculate the force acting on his knee joints. Note that since Andrew steps off the platform, his initial y-velocity is 0 m/s. Considering that 1000 N is roughly equal to 225 lbs, do you think this hurts Andrew's knees?)

ANSWER: 598.92 kN

when he first makes contact with the ground, his velocity is at a maximum:
his energy is transformed as follows: $PE_g \rightarrow KE \rightarrow PE_s \rightarrow 0$
His velocity rapidly drops to 0 when his knee joints bear the force.
Use this with the work-energy theorem. $W_{net} = \Delta KE$

$$v^2 = v_i^2 + 2a\Delta y$$

$$v = \sqrt{2(-9.8 \frac{m}{s^2})(-3.1m)}$$

$$v = 7.7949 \frac{m}{s}$$

$$F_{net} d \cos \theta = \frac{1}{2} m v_f^2 - \frac{1}{2} m v_i^2$$

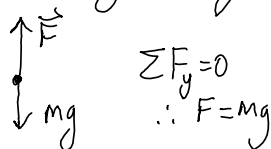
$$f_k \Delta y_c \cos 180^\circ = 0 - \frac{1}{2} m (2a_y \Delta y)$$

$$f_k = m a_y \frac{\Delta y}{\Delta y_c} = (96.6 \text{ kg})(-9.8 \frac{m}{s^2}) \left(\frac{-3.1m}{0.49 \times 10^{-2} m} \right) = 598920 \text{ N}$$

8. Julia lifts a 42-kg barbell from the ground to a height of 2.0 m. How much work was done on the barbell?

ANSWER: 823 J

work against gravitational field



$$W = F d \cos \theta_{Fd}$$

$$W = mgh \cos 0$$

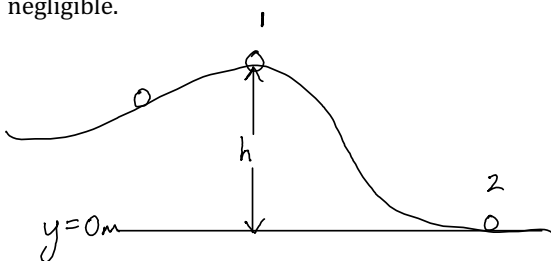
$$W = mgh$$

$$W = (42 \text{ kg})(9.8 \frac{m}{s^2})(2 \text{ m})$$

$$W = 823 \text{ J}$$

9. Thomas is riding on a roller coaster, ascending a steep track as he approaches the top of a 79.0-m peak. Assuming Thomas is at rest at the top of the peak, what is his speed when he reaches level ground at 0 m? Assume that friction is negligible.

ANSWER: 39.35 m/s



no friction
no externally applied forces.
∴ no nonconservative forces
∴ $W_{nc} = 0$
∴ $\Delta E_t = 0$ ($\because \Delta E_t = W_{nc}$)

$$\Delta E_t = 0$$

$$E_2 - E_1 = 0$$

$$E_1 = E_2$$

$$mgh = \frac{1}{2} m v^2$$

$$v = \sqrt{2gh}$$

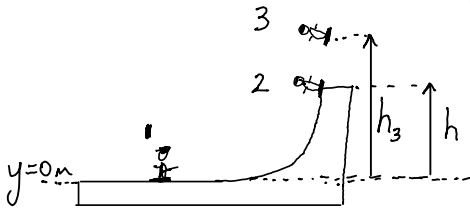
$$v = \sqrt{2(9.8 \frac{m}{s^2})(79m)}$$

$$v = 39.35 \frac{m}{s}$$

10. Justin skateboards up a curved ramp (quarterpipe) and launches himself into the air. He begins his motion with a speed of 11.6 m/s at a total distance of 4.0 m away from the launch position. In order to launch himself from the top of the quarterpipe, Justin does 290 J of work. Justin has a mass of 87 kg and knows that the average friction force acting on him as he skates across this particular surface is 12 N. If the quarterpipe has a height of 0.5 m, how far above the quarterpipe does Justin reach?

find $h_{\text{above}} = h_3 - h$

ANSWER: 6.65 m



$$E_1 = KE$$

$$E_2 = KE + PE_g$$

$$E_3 = PE_g$$

d_{12} = distance traveled from 1 → 2 = 4.0 m

Several ways to solve this. Easiest is presented.

$$\Delta E_b = W_{nc}$$

$$E_3 - E_1 = W_{\text{friction}} + W_{\text{Justin}}$$

$$mgh_3 - \frac{1}{2}mv_1^2 = f d_{12} \cos \theta_{fd} + W_{\text{Justin}}$$

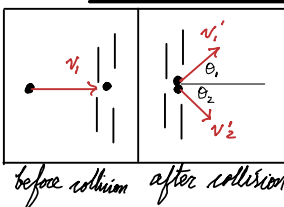
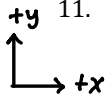
$$h_3 = \frac{1}{mg} (f d_{12} \cos \theta_{fd} + W_{\text{Justin}} + \frac{1}{2}mv_1^2)$$

$$h_{\text{above}} = \frac{1}{mg} (f d_{12} \cos \theta_{fd} + W_{\text{Justin}} + \frac{1}{2}mv_1^2) - h$$

$$h_{\text{above}} = \frac{1}{(87 \text{ kg})(9.8 \text{ m/s}^2)} [(12 \text{ N})(4 \text{ m})(\cos 180^\circ) + (290 \text{ J}) + \frac{1}{2}(87 \text{ kg})(11.6 \text{ m/s})^2] - 0.5 \text{ m} = \underline{6.649 \text{ m}}$$

(W_{friction} is work done by friction.)
(W_{Justin} is work done by Justin.)

11. A 0.24-kg yellow box is slid eastward on a frictionless surface into a dark room, where it strikes an initially stationary green box, which has a mass of 0.18 kg. The yellow box emerges from the room at an angle of 25° north of east. The speed of the yellow box is originally 8.9 m/s and is 7.1 m/s after the collision. Calculate the angle at which the green box travels after the collision.



two boxes: ① yellow ② green

find θ_z

ANSWER: 50.6° S of E

$$\Delta \vec{p}_x = 0$$

$$\vec{p}_x' - \vec{p}_x = 0$$

$$\vec{p}_x' = \vec{p}_x$$

$$m_1 v_1 \cos \theta_1 + m_2 v_2' \cos \theta_2 = m_1 v_1$$

$$\Delta \vec{p}_y = 0$$

$$\vec{p}_y' - \vec{p}_y = 0$$

$$\vec{p}_y' = \vec{p}_y$$

$$m_1 v_1 \sin \theta_1 + m_2 (-v_2' \sin \theta_2) = 0$$

$$\theta_z = \tan^{-1} \left(\frac{7.1 \text{ m/s} \sin 25^\circ}{8.9 \text{ m/s} - 7.1 \text{ m/s} \cos 25^\circ} \right)$$

$$\theta_z = 50.594^\circ \text{ South of east}$$

for all collisions, $\Delta \vec{p} = 0$
we do not assume this is an elastic collision

$$\Delta \vec{p} = 0$$

$$\Delta (\vec{p}_x + \vec{p}_y) = 0$$

$\Delta \vec{p}_x + \Delta \vec{p}_y = 0$ this means both $\Delta \vec{p}_x = 0$ & $\Delta \vec{p}_y = 0$ since they are orthogonal vectors

Choose your preferred way to solve a system of equations

$$m_2 v_2' \cos \theta_2 = m_1 v_1 - m_1 v_1' \cos \theta_1 \quad m_1 v_1' \sin \theta_1 = m_2 v_2' \sin \theta_2$$

$$v_2' = \frac{m_1 v_1 - m_1 v_1' \cos \theta_1}{m_2 \cos \theta_2} = v_2' = \frac{m_1 v_1' \sin \theta_1}{m_2 \sin \theta_2}$$

Simplify & solve for θ_2

$$(m_1 v_1 - m_1 v_1' \cos \theta_1) (m_2 \sin \theta_2) = (m_2 \cos \theta_2) (m_1 v_1' \sin \theta_1)$$

$$\frac{\sin \theta_2}{\cos \theta_2} = \tan \theta_2 = \frac{m_1 v_1' \sin \theta_1}{m_1 v_1 - m_1 v_1' \cos \theta_1}$$

$$\theta_2 = \tan^{-1} \left(\frac{m_1 v_1' \sin \theta_1}{m_1 v_1 - m_1 v_1' \cos \theta_1} \right) = \tan^{-1} \left(\frac{v_1' \sin \theta_1}{v_1 - v_1' \cos \theta_1} \right)$$

12. Mary is a 87-kg athlete competing in a race. She begins from rest with an average acceleration of 1.8 m/s² for 1.7 s. What is her momentum at 1.7 s?

ANSWER: 266.22 $\frac{\text{kg} \cdot \text{m}}{\text{s}}$

find $|\vec{p}(t=1.7 \text{ s})| = p(t=1.7 \text{ s})$

$$p = mv \therefore p(t) = mv(t)$$

We must find her velocity at 1.7 s using kinematics (easiest way)

$$\vec{v}(t) = \vec{v}_0 + \vec{a}t$$

$$\therefore \vec{p}(t) = m(\vec{v}_0 + \vec{a}t)$$

$$p(t=1.7 \text{ s}) = (87 \text{ kg})(0 \text{ m/s} + (1.8 \text{ m/s}^2)(1.7 \text{ s})) = \underline{266.22 \frac{\text{kg} \cdot \text{m}}{\text{s}}}$$